MOULI VUNNAM

Michigan, USA (Open to relocate) | vunnammouli9@gmail.com | (616)-232-6957 | https://www.linkedin.com/in/mouli-vunnam/

SUMMARY

Data and Analytics Engineer with 4+ years of experience building production-grade ETL pipelines, data models, and analytics infrastructure across fintech, education, and early-stage products. Delivered over $1M in efficiency and cost savings through scalable pipelines built with Snowflake, dbt, SQL, and Python. Proven track record in anomaly detection, data quality validation, forecasting accuracy, and enabling engineering and product teams with reliable, trustworthy data systems.

SKILLS

Programming and Analytics: Python, SQL, Pandas, NumPy, Shell Scripting, scikit-learn
Databases and SQL: Snowflake, BigQuery, PostgreSQL, MySQL, SQL Server, Advanced SQL, Window Functions, Query Optimization, Dimensional Modeling
Data Engineering and Cloud: ETL/ELT, dbt, Airflow, Spark, API Integration, Databricks, Azure, AWS, GCP, Docker, Kubernetes, CI/CD, Terraform, Data Quality Validation, Data Reliability Monitoring, Streaming Fundamentals
ML and Modeling: Feature Engineering, Regression, Classification, Clustering, Anomaly Detection, Trend Analysis, Model Evaluation, Drift Monitoring, MLflow, TensorFlow, PyTorch
Visualization and Tools: Tableau, Power BI, Matplotlib, Seaborn, KPI Design, A/B Testing, Git, Linux, Jupyter, RESTful APIs, Jira, Excel

PROFESSIONAL EXPERIENCE

Data Analyst ***Chicago Education Advocacy Cooperative, Chicago, USA*** **June 2025 - Present**

- Automated ETL pipelines ingesting attendance and performance data from 6 partner schools, reducing weekly manual cleanup from ~6 hours to under 1 hour and enabling near-hourly data refreshes.
- Deployed containerized ETL workflows using Docker and Kubernetes, with infrastructure provisioned via Terraform and automated through CI/CD pipelines.
- Designed Snowflake schemas and dbt data models that reduced reporting rework by 30% and enabled new KPI tracking across student engagement and intervention programs.
- Built dashboards tracking intervention impact and student progress, directly improving how leadership teams allocated resources.
- Analyzed behavioral and attendance trends to surface early dropout risk signals, improving early identification accuracy by ~12% over manual checks.

Data Analyst ***Stripe, Chicago, USA*** **May 2024 – May 2025**

- Built and maintained SQL pipelines processing 4–6 million daily transactions: reduced reporting lag from 36 hours to ~6 hours for BI and leadership teams.
- Analyzed transaction volume patterns, customer behavior, and conversion trends to identify anomalies and explain unexpected drops to product and engineering teams.
- Partnered with engineers to redefine event tracking schemas and metric logic, improving clean event coverage by ~18%.
- Designed and delivered dashboards used by leadership to measure feature launch performance, adoption trends, and payment funnel health.
- Improved forecasting stability by refining feature sets, monitoring model drift, and validating expected vs. actual metrics over time using MLflow.
- Collaborated with product managers, engineers, and business stakeholders to define metrics, validate data across sources, and align reporting requirements.

Data Engineer ***FIS, Chennai, India*** **Jan 2023 – Aug 2023**

- Built ETL workflows migrating data from legacy systems into analytics-ready datasets and data marts, supporting millions of records per batch run.
- Optimized SQL queries powering business-critical reports, reducing runtime from minutes to seconds and significantly lowering compute costs.
- Delivered clean, validated feature datasets used in fraud detection models and downstream risk analysis pipelines.
- Partnered with product and compliance teams to define data requirements for risk analysis, audit readiness, and regulatory reporting.
- Created pipeline monitoring scripts and documentation that improved reliability and reduced production failures.

Data Science Intern ***1Stop.ai, Chennai, India*** **Nov 2020 – Nov 2021**

- Cleaned and transformed raw datasets for ML experiments; built baseline classification and clustering models that fed into production workflows.
- Automated routine preprocessing steps using Python, saving significant manual effort for the data team.
- Contributed to exploratory data analysis that informed model selection and feature prioritization for production ML pipelines.

EDUCATION

MS in Computer Science **Central Michigan University, Mount Pleasant** **GPA: 3.78/4.0**